

Jatin Jain

Rochester, NY

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EDUCATION

Rochester Institute of Technology

May 2028

Bachelor of Science in Computer Science

Minor in Computer Engineering & Immersion in Mathematics

- **RIT Founders Scholarship for International Students – Merit-based scholarship for excellence**
- **of Computing - Dean's List – Fall 2023, 2024**

Systems & Design: Digital Systems & Design, Intro to AI, Computer Science Theory, Parallel and Distributed Systems (C#, Threading, Networking), Concepts of Computer Systems (Assembly), Robot-Operating Systems

Programming Languages: Python, Java, JavaScript, C, C++, Kotlin, C#, SQL, Angular, Node.js, Java Spring, REST API, Flask, FastAPI, OpenCV, PyTorch, TensorFlow, CMake, ARM Tools

Environments: Unity, OpenGL, Meta Quest SDK, Blender, XR Interaction Toolkit

Certifications: **NVIDIA** Accelerated Computing with Modern CUDA C++, **MIT Institute for Data, Systems, and Society (IDSS):** Data Science and Machine Learning

Experience

Software Engineer Co-op | Eoptic Inc.

Aug 2025 – now

Technologies: C++, ROS2, Machine Learning

- **Engineered core C++ libraries** for a Pan-Tilt Unit in a **multi-million-dollar optics project**, ensuring seamless integration into a ROS2 robotics framework.
- **Developed and tested embedded firmware under extreme environmental conditions** (thermal, vibration, and stress) to identify system weak points and strengthen reliability.
- **Designed calibration routines and contributed to an ML-driven camera alignment pipeline**, supporting high-precision imaging and sensor synchronization.
- **Owned debugging and validation workflows across hardware platforms**, producing robust documentation and test protocols adopted by the engineering team.

Research Assistant | School of Mathematics & Statistics, National Institutes of Health

Technologies: FastAPI, Flask, Python, NumPy, Matplotlib, TypeScript, react

May 2025 – now

- Enhanced NIH-funded Type-2 Diabetes simulation by adding 6+ metabolic variables and 10+ liver reactions, improving accuracy ~40%.
- Optimized Python differential equation solvers, cutting runtime by **90% (30s → <3s)**.
- Built and deployed 3 FastAPI endpoints + Flask frontend, enabling **25+ personalized simulations** for researchers.
- Produced 15+ scientific plots for NIH progress reports and manuscript submissions; managed Git version control (500+ commits). Contributed to grant deliverables under **NIH Award #R16GM154782**

Tutor | Academic Success Centre, HEOP Program

Jan 2024 – Present

- Mentored 1700+ students in STEM courses through personalized sessions and collaborative workshops to enhance academic performance.

ACTIVITIES, CLUBS, ORGANISATIONS

Electric Vehicle Team | RIT Firmware Team

Nov 2023 – Present

- **Engineered embedded firmware for APDB board using C++ and ARM protocols, improving sensor data throughput by 30%**

OASIS | RIT Executive Board: Vice-President

Aug 2024 – Present

- Spearheaded operations for a 300+ member cultural organisation, coordinating logistics, sponsorships for **8+ major events(every academic year)** including Diwali and Holi.

PROJECTS

Cyber Threat Intelligence Dashboard | Python, Flask, MITRE ATT&CK, OTX

May 2025

- Built a full-stack dashboard aggregating threat intel from AlienVault OTX.
- Automated IOC extraction and MITRE ATT&CK mapping via TAXII/STIX feeds.

Real-Time ASL Interpreter | Python, OpenCV, RESNET-18

Dec 2024

- Developed a real-time ASL recognition system (95% accuracy on 50K+ samples).
- Optimized on-device inference, cutting latency 30% vs. cloud models.